Georgia Trout Stream Restoration: Site & Watershed Database

Blue Ridge Stream Restoration & Mitigation LLC

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Ultimate KPI: Fund world-class fly fishing trip to Argentina!

To help Hunter scale beyond his flagship showcases at **Anderson Creek (Roya's Cabin)** and **Goldmine Hollow (Hunter's Cabin)**, this database maps out **10 high-priority trout watersheds in North Georgia** where stream degradation is severe and opportunities for stream mitigation banking, Permittee-Responsible Mitigation (PRM), or high-end private landowner restorations are highly abundant.

■■ High-Priority Watershed Reaches & Sourcing Map

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                     NORTH GEORGIA TROUT WATER BASINS

      [ WEST: COOSA SYSTEM ]                   [ EAST: CHATTAHOOCHEE SYSTEM ]
* Toccoa River (Fannin County)         * Soque River (Habersham County)
* Noontootla Creek (Fannin County)     * Nacoochee Valley / Chattahoochee (White Co)
* Cartecay River (Gilmer County)       * Upper Chestatee River (Lumpkin County)
* Ellijay River (Gilmer County)        * Tallulah River Tributaries (Rabun County)
* Fightingtown Creek (Fannin County)   * Cooper Creek Basin (Union County)
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1. Toccoa River Watershed (Fannin County)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Premium designated coldwater trout waters (Brown, Rainbow, and Brook trout).
- **Fluvial Degradation Profile**:
- *Siltation*: High-velocity runoff from unpaved logging and cabin access roads (e.g., Doublehead Gap Road, Newport Road, and Peter Knob Road) deposits massive sand bars into tributaries, smothering spawning gravels.
- *Bank Erosion*: Heavy residential development along the river banks has led to bank clearing and active lateral failures.
- **Mitigation & Restoration Opportunity**:
- *Private Showroom Reaches*: High-net-worth estate properties along the upper river (from Dial to Deep Gap) require geomorphic log structures and rock vanes to protect actively eroding lawn boundaries.
- *PRM Sites*: Commercial developers looking for Coosa basin offsets can acquire 10–20 acre degraded riparian buffer lots to execute rapid restorations under NWP 27.

2. Noontootla Creek Basin (Fannin & Gilmer Counties)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Legendary wild trout fishery with strict artificial-only state regulations.
- **Fluvial Degradation Profile**:
- *Severe Sand Loading*: Heavy erosion from unpaved dirt roads (specifically U.S. Forest Service Road 58) acts as a constant conveyor belt of fine granite sand, filling deep holding pools and smothering benthic macroinvertebrates.
- *Headcutting*: Small mountain headwater tributaries are experiencing severe headcuts, migrating upstream and washing out forest soils.
- **Mitigation & Restoration Opportunity**:
- *Cedar Log Step-Pools*: Replicate the **Goldmine Hollow Showcase** model. Property owners along Noontootla Creek tributaries will pay a premium for natural drop structures that trap upstream road sand before it smothers main-stem spawning beds.
- *US Forest Service Partnerships*: Potential for design-build subcontracts to install geomorphic sediment basins along FS-58.

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3. Soque River Basin (Habersham County)

- **Basin/HUC**: Upper Chattahoochee Basin (HUC `03130001`)
- **Trout Status**: Nationally famous private trophy trout river (brook/brown/rainbow trout reaching massive sizes).
- **Fluvial Degradation Profile**:
- *Agricultural Widening*: Historic cattle grazing and pasture clearing removed native deep-rooted riparian vegetation, allowing the river to widen from a stable $18\text{-foot} *channel to a shallow, slow $35\text{-to-}45\text{-foot}* sheet.
- *Solar Warming*: Unshaded pasture sections easily exceed 68°F in mid-summer, threatening trout survival.
- **Mitigation & Restoration Opportunity**:
- *High-End Private Restoration (Fee-for-Service)*: Landowners on the Soque charge up to $500/day per rod for trophy trout angling. They are highly motivated to fund private geomorphic meander restorations (Rosgen Class C4) to build deep holding pools and fast aerated riffles.
- *Commercial Banks*: Excellent watershed for large pasture acquisitions to convert into dedicated stream mitigation banks.

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4. Cartecay River Watershed (Gilmer County / Ellijay)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Designated trout waters; high recreational tubing and paddling pressure.

- **Fluvial Degradation Profile**:
- *Impervious Runoff*: Intense construction of mountain cabins on steep slopes has led to high-velocity stormwater runoff.
- *Lateral Bank Collapse*: The river banks consist of highly erodible alluvial soils. Increased peak flows cause massive blocks of riverbank to shear off into the channel.
- **Mitigation & Restoration Opportunity**:
- *Root Wad Bioengineering*: Showcase how root wads (using cedar or oak root structures harvested on-site) can be embedded into eroding outer bends. This provides immediate, natural structural support while creating outstanding cover for large holding trout.
- *PRMs*: Developers can offset commercial impacts in Gilmer/Cherokee counties by executing restorations along degraded Cartecay agricultural tributaries.

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5. Ellijay River Watershed (Gilmer County)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Designated trout water; local headwaters are highly productive spawning streams.
- **Fluvial Degradation Profile**:
- *Pasture Runoff*: Unfenced livestock access has trampled banks, leading to severe lateral bank erosion and high bacteria loading.
- *Overwidened Channels*: Reaches flowing through active farmlands have lost their deep geomorphic cross-sections, resulting in silted-in pools and flat runs.
- **Mitigation & Restoration Opportunity**:
- *Priority 1 Reconnection*: The floodplain in Ellijay's valley regions is highly accessible. Hunter can execute Priority 1 geomorphic designs—building a completely new, stable, meandering channel on the historic floodplain and filling the old, degraded ditch.
- *USACE Multipliers*: Priority 1 designs yield the absolute highest credit multiplier (up to **8.6**) under the Savannah District SOP.

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6. Fightingtown Creek (Fannin County)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Designated premium wild trout stream; highly sought-after private fly fishing water.
- **Fluvial Degradation Profile**:
- *High-Gradient Bank Stress*: Steep channel slopes create massive hydraulic shear stress against stream banks during storm events. Uninformed bank modifications (like dumping rip-rap) have worsened downstream velocity.
- *Loss of Canopy Cover*: Rapid residential cabin development has led to buffer clearing, exposing high-gradient waters to direct solar warming.
- **Mitigation & Restoration Opportunity**:

- *Boulder Cross-Vanes & J-Hooks*: Install native rock cross-vanes to redirect central flow, scour out stable mid-channel pools, and keep water temperatures cool.
- *Buffer Re-establishment Contracts*: Planting native Eastern Hemlock and Mountain Laurel stands to satisfy local county environmental buffer variance rules.

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7. Nacoochee Valley / Upper Chattahoochee River (White County)

- **Basin/HUC**: Upper Chattahoochee Basin (HUC `03130001`)
- **Trout Status**: Prime historic trout fishing valley.
- **Fluvial Degradation Profile**:
- *Valley Siltation*: Centuries of agricultural tillage have deposited thick layers of clay and fine sand over the river’s historic gravel-cobble substrate.
- *Solar solar heating*: Wide, slow, unshaded channels running through valley pastures experience significant warming.
- **Mitigation & Restoration Opportunity**:
- *Channel Narrowing & Meander Restoration*: Using bioengineered log vanes to construct stable inner-bank benches (floodplain benches) that physically narrow the river bankfull width, accelerating water speed to flush out fine clay deposits.
- *High-Yield Credit Supply*: Sells credits directly to developers in Hall, Gwinnett, and Forsyth counties who are actively piping Piedmont tributaries.

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8. Cooper Creek Basin (Union County)

- **Basin/HUC**: Coosa River Basin (HUC `03150102`)
- **Trout Status**: Designated trout waters; popular state stocking and wild trout system.
- **Fluvial Degradation Profile**:
- *Unpaved Road Sediment*: Forest Service roads parallel the creek for miles, shedding fine sands and choking active gravel redds.
- *Loss of Wood Structure*: Historic logging removed instream woody debris (logs, root wads), leaving the stream lacking natural pool-scouring structures.
- **Mitigation & Restoration Opportunity**:
- *Instream Wood Placement*: High opportunity to execute "Large Woody Debris" (LWD) configurations using native hemlock/pine logs, which physically scour pools and provide immediate organic carbon lift.
- *State EPD / USACE Grants*: Potential for Section 319 Clean Water Act grants to fund watershed sediment reductions.

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9. Upper Chestatee River Watershed (Lumpkin County)

- **Basin/HUC**: Upper Chattahoochee Basin (HUC `03130001`)

- **Trout Status**: Designated trout waters; high-gradient mountain tributaries.
- **Fluvial Degradation Profile**:
- *Gold Mining Erosion Residuals*: Historic instream mining activities severely altered channel paths, leaving behind unstable, eroding sand banks and tailing piles.
- *Stormwater Energy*: Development around Dahlonega causes rapid hydrologic spikes that widen headwater streams.
- **Mitigation & Restoration Opportunity**:
- *PRM Offsets*: Lumpkin County is a primary target for exurban residential developments. Land developers facing local stream crossings represent a direct target for Permittee-Responsible restorations.

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10. Tallulah River Tributaries (Rabun County)

- **Basin/HUC**: Savannah River Basin (HUC `03060102`)
- **Trout Status**: Highly productive wild trout waters; high-gradient brook trout headwaters.
- **Fluvial Degradation Profile**:
- *Headwater Channel Degradation*: Road and trail erosion dump granite silt into spawning beds, blocking natural egg oxygenation.
- *Steep Bank Undercutting*: High energy storm flows undercut stream banks, causing trees to fall prematurely and blocking fish passage.
- **Mitigation & Restoration Opportunity**:
- *High-Gradient Step-Pool Systems*: Construct stable boulder and cedar log step-pool cascades to trap sediment and create critical juvenile rearing habitats.
- *Savannah Basin Offsets*: Restoring streams here generates premium Savannah River Basin stream credits, highly sought-after for Northeast Georgia transportation expansions.